Exercise 13.1

Function of two variables

A function of two variables x & y is a rule that assigns a unique real numbers $f(x,y)$ to each point (x,y) in some set D in xy-plane

Question no 1

$f(x,y) = x^2y + 1$

(a) $f(2,1)$
(b) $f(1,2)$
(c) $f(0,0)$
(d) $f(1,-2)$
(e) $f(4a, a)$
(f) $f(ab, a-b)$

a) $x^2y + 1$
$(2)^2(1) + 1$
$= 4+1 = 5$

b) $x^2 + 1$
$(1)^2(2) + 1$
$= 3$

(c) $x^2 + 1$
$0 + 1$
$= 1$

d) $x^2y + 1$
$= (1)^2(-2) + 1$
$= -1$

e) $x^2y + 1$
$(4a)^2(a) + 1$
$= 16a^3 + 1$

f) $x^2y + 1$
$(ab)^2(a-b) + 1$
$a^3b^2 - a^2b^3 + 1$

Question 2

$f(x,y) = x + \sqrt[3]{xy}$

a) $f(t, t^5)$
b) $f(x, x^5)$
c) $f(2y^2, 4y)$

a) $x + \sqrt[3]{xy}$
$t + \sqrt[3]{t(t^5)}$
$t + t^{6 \times 1/3}$
$t + t^2$

b) $x + \sqrt[3]{xy}$
$x + \sqrt[3]{x^6}$
$x + x^2$
$x(1+x)$

c) $2y^2 + \sqrt[3]{2y^2 \times 4y}$
$2y^2 + \sqrt{8y^3}$
$2y^2 + 2y$
$2y(y+1)$

Question 3

$f(x,y) = xy + 3$

a) $f(x+y, x-y)$

b) $f(xy, 3x^2y^4)$

a) $(x+y)(x-y)+3$

x^2-y^2+3

b) $(xy)(3x^2y^4)+3$

$3x^3y^5+3$

Question 4

$g(x) = x \sin x$

a) $g\left(\frac{x}{y}\right)$

$= \frac{x}{y} \sin\left(\frac{x}{y}\right)$

b) $g(xy^2)$

$xy^2 \sin xy^2$

c) $g(x-y)$

$(x-y)\sin(x-y)$

Question 5

Find $f(g(x), h(y))$ if $f(x,y) = xe^{xy}$

$g(x) = x^3$

$h(y) = 3y+1$

Solution

$$f(x,y) = xe^{xy}$$
$$= g(x)\, e^{g(x)h(y)}$$
$$= x^3 e^{(x^3)(3y+1)} \text{ — ANS}$$

Question 6

Find $g(u(x,y), v(x,y))$ if $g(x,y) = y\sin(x^2y)$

$u(x,y) = x^2y^3$

$v(x,y) = \pi xy$

Solution

$$g(x,y) = y\sin(x^2y)$$

$$= v(x,y)\sin((u(x,y))^2 v(x,y))$$
$$= \pi xy \sin((x^2y^3)^2(\pi xy))$$
$$= \pi xy \sin((x^4y^6)(\pi xy))$$
$$= \pi xy \sin(\pi x^5y^7) \text{ — ANS}$$

Question 7

Let $f(x,y) = x + 3x^2y^2$

$x(t) = t^2$

$y(t) = t^3$

a) $f(x(t), y(t))$

$f(x,y) = x + 3x^2y^2$

$= t^2 + 3\,(t^2)^2(t^3)^2$

$= t^2 + 3(t^4)(t^6)$

$= t^2 + 3t^{10}$ — ANS

b) $f(x(0), y(0))$

$f(x,y) = x + 3x^2y^2$

$= 0 + 3(0)(0)$

$= 0$ — ANS

c) $f(x(2), y(2))$

$f(x,y) = x + 3x^2y^2$

$= 2^2 + 3(2^2)^2(2^3)^2$

$= 2^2 + 3(4)^2(8)^2$

$= 2^2 + 3072$

$= 3076$ — ANS

Question 8

let $g(x,y) = ye^{-3x}$

$x(t) = \ln(t^2+1)$

$y(t) = \sqrt{t}$

$g(x(t), y(t))$

$g(x,y) = ye^{-3x}$

$= \sqrt{t}\,e^{-3(\ln(t^2+1))}$ — ANS

Exercise 13.2

Use limit laws and continuity properties to evaluate the limit.

Question 1

$$\lim_{(x,y)\to(1,3)} (4xy^2 - x)$$

$$= 4(1)(3)^2 - 1$$

$$= 35 \quad \text{— ANS}$$

Question 2

$$\lim_{(x,y)\to(0,0)} \frac{4x-y}{\sin y - 1}$$

$$= \frac{4(0) - 0}{\sin(0) - 1}$$

$$= 0 \quad \text{— ANS}$$

Question 3

$$\lim_{(x,y)\to(-1,2)} \frac{xy^3}{x+y}$$

$$= \frac{(-1)(2)^3}{-1+2}$$

$$= \frac{-8}{1}$$

$$= -8 \quad \text{— ANS}$$

Question 4

$$\lim_{(x,y)\to(1,-3)} e^{2x-y^2}$$

$$= e^{2-(-3)^2}$$

$$= e^{2-9}$$

$$= e^{-7} \quad \text{— ANS}$$

Question 5

$$\lim_{(x,y)\to(0,0)} \ln(1+x^2y^3)$$

$$= \ln(1+(0)(0)$$

$$= \ln 1$$

$$= 0 \text{ — ANS}$$

Question 6

$$\lim_{(x,y)\to(4,-2)} x\sqrt[3]{y^3+2x}$$

$$= 4\sqrt[3]{(-2)^3+2(4)}$$

$$= 4\sqrt[3]{-8+8}$$

$$= 4(0)$$

$$= 0 \text{ — ANS}$$

Determine wether the limit exists if so find its value

Question 13

$$\lim_{(x,y)\to(0,0)} \frac{x^4-y^4}{x^2+y^2} \quad - \quad \frac{0}{0} \text{ form}$$

$$= \frac{(x^2)^2-(y^2)^2}{x^2+y^2}$$

$$= \frac{\cancel{(x^2+y^2)}(x^2-y^2)}{\cancel{x^2+y^2}}$$

$$= (x^2-y^2)$$

$$= (0^2-0^2)$$

$$= 0 \text{ — ANS}$$

Question 14

$$\lim_{(x,y)\to(0,0)} \frac{x^4-16y^4}{x^2+4y^2} \quad - \frac{0}{0} \text{ form}$$

$$= \frac{(x^2)^2-(4y^2)^2}{x^2+4y^2}$$

$$= \frac{\cancel{(x^2+4y^2)}(x^2-4y^2)}{\cancel{(x^2+4y^2)}}$$

$$= x^2-4y^2$$

$$= 0-0$$

$$= 0 \text{ — ANS}$$

Question 17

$$\lim_{\underset{z}{(x,y)}\to(2,-1,2)} \frac{xz^2}{\sqrt{x^2+y^2+z^2}}$$

$$= \frac{(2)(2)^2}{\sqrt{2^2+(-1)^2+2^2}}$$

$$= \frac{8}{\sqrt{4+1+4}}$$

$$= \frac{8}{\sqrt{9}}$$

$$= \frac{8}{3} \text{ — ANS}$$

Exercise 13.3

Partial deriviative

Example 2

$f(x,y) = 2x^3y^2 + 2y + 4x$

find $f_x(x,y)$

$f_y(x,y)$

WRT X

$f_x(x,y) = \frac{\partial}{\partial x}\left(2x^3y^2 + 2y + 4x\right)$

$= 6x^2y^2 + 4$ — ANS

WRT Y

$f_y(x,y) = \frac{\partial}{\partial y}\left(2x^3y^2 + 2y + 4x\right)$

$= 4x^3y + 2$ — ANS

Example 3

$f(x,y) = x^4\sin(xy^3)$

WRT X

$f_x(x,y) = x^4\frac{\partial}{\partial x}\sin(xy^3) + \sin(xy^3)\frac{\partial}{\partial x}x^4$

$= x^4(\cos(xy^3))(y^3) + \sin(xy^3)\,4x^3$

$= x^4y^3\cos(xy^3) + 4x^3\sin(xy^3)$ — ANS

WRT Y

$f_y(x,y) = \frac{\partial}{\partial y}x^4\sin(xy^3)$

$= x^4\cos(xy^3)(3y^2x)$

$= 3x^5y^2\cos(xy^3)$ — ANS

Example 9

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

WRT X

$$f_x(x,y) = -\frac{(x^2+y^2)(y) - (xy)(2x)}{(x^2+y^2)^2}$$

$$= -\frac{x^2y + y^3 - 2x^2y}{(x^2+y^2)^2}$$

$$= \frac{x^2y - y^3}{(x^2+y^2)^2} \quad \text{— ANS}$$

WRT y

$$f_y(x,y) = -\frac{(x^2+y^2)(x) - (xy)(2y)}{(x^2+y^2)^2}$$

$$= -\frac{x^3 + xy^2 - 2xy^2}{(x^2+y^2)^2}$$

$$= \frac{xy^2 - x^3}{(x^2+y^2)^2} \quad \text{— ANS}$$

Example 10

if $f(x,y,z) = x^3y^2z^4 + 2xy + z$ then

WRT X

$= f_x(x,y,z) = 3x^2y^2z^4 + 2y$ — ANS

WRT y

$= f_y(x,y,z) = 2x^3yz^4 + 2x$ — ANS

WRT z

$= f_z(x,y,z) = 4x^3y^2z^3 + 1$ — ANS

Example 11

if $f(\rho, \theta, \phi) = \rho^2 \cos\phi \sin\theta$

WRT ρ

$f_\rho(\rho, \theta, \phi) = 2\rho \cos\phi \sin\theta$

WRT θ

$f_\theta(\rho, \theta, \phi) = \rho^2 \cos\phi \cos\theta$

WRT ϕ

$f_\phi(\rho, \theta, \phi) = -\rho^2 \sin\phi \sin\theta$

Example 12

Second order partial derivichive

$f(x,y) = x^2y^3 + x^4y$

$f_x(x,y) = 2xy^3 + 4x^3y = \partial f/\partial x$

$f_y(x,y) = 3x^2y^2 + x^4 = \partial f/\partial y$

$f_{x^2}(x,y) = 2y^3 + 12x^2y = \partial^2 f/\partial x^2$

$f_{y^2}(x,y) = 6x^2y = \partial^2 f/\partial y^2$

$f_{xy}(x,y) = 6xy^2 + 4x^3 = \partial f/\partial y\partial x$

$f_{yx}(x,y) = 6xy^2 + 4x^3 = \partial f/\partial x\partial y$

Example 13

$f(x,y) = y^2e^x + y$ find f_{xyy}

$f_x(x,y) = y^2e^x \quad . \quad \partial f/\partial x$

$f_{xy}(x,y) = 2ye^x \quad . \quad \partial^2 f/\partial y \partial x$

$f_{xyy}(x,y) = 2e^x \quad = \quad \partial^3 f/\partial y^2 \partial x$ — ANS

Question 25

Find $\frac{\partial z}{\partial x}$ & $\frac{\partial z}{\partial y}$

$z = 4e^{x^2y^3}$

WRT X

$\frac{\partial z}{\partial x} = 4e^{x^2y^3} \cdot 2xy^3$

$= 8xy^3e^{x^2y^3}$ — ANS

WRT Y

$\frac{\partial z}{\partial y} = 4e^{x^2y^3} \cdot 3x^2y^2$

$= 12x^2y^2\, 4e^{x^2y^3}$ — ANS

Question 26

$z = \cos(x^5y^4)$

WRT X

$\frac{\partial z}{\partial x} = -\sin(x^5y^4) \cdot 5x^4y^4$

$= -5x^4y^4 \sin(x^5y^4)$ — ANS

WRT Y

$\frac{\partial z}{\partial y} = -\sin(x^5y^4) \cdot 4x^5y^3$

$= -4x^5y^3 \sin(x^5y^4)$ — ANS

Question 28

$$z = e^{xy} \sin 4y^2$$

Wrt x

$$\frac{\partial z}{\partial x} = e^{xy} \sin 4y^2 \,(y)$$

$$= y e^{xy} \sin(4y^2) \quad \text{— ANS}$$

Wrt y

$$\frac{\partial z}{\partial y} = e^{xy} \frac{\partial}{\partial y} \sin 4y^2 + \sin 4y^2 \frac{\partial}{\partial y} e^{xy}$$

$$= e^{xy} \cdot \cos(4y^2)(8y) + \sin 4y^2 \cdot e^{xy} \cdot (x)$$

$$= 8y e^{xy} \cos(4y^2) + x e^{xy} \sin(4y^2) \quad \text{— ANS}$$

Question 29

$$z = \frac{xy}{x^2 + y^2}$$

WRT x

$$\frac{\partial z}{\partial x} = \frac{(x^2+y^2)\,\partial/\partial x\,(xy) - (xy)\,\partial/\partial x\,(x^2+y^2)}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2)(y) - (xy)(2x)}{(x^2+y^2)^2}$$

$$= \frac{x^2y + y^3 - 2x^2y}{(x^2+y^2)^2}$$

$$= \frac{y^3 - x^2y}{(x^2+y^2)^2}$$

$$= -\frac{y(x^2-y^2)}{(x^2+y^2)^2} \quad \text{— ANS}$$

WRT y

$$\frac{\partial z}{\partial y} = \frac{(x^2+y^2)\,\partial/\partial y\,(xy) - (x^2y)\,\partial/\partial y\,(x^2+y^2)}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2)(x) - (xy)(2y)}{(x^2+y^2)^2}$$

$$= \frac{x^3+xy^2-2xy^2}{(x^2+y^2)^2}$$

$$= \frac{x^3-xy^2}{(x^2+y^2)^2}$$

$$= \frac{x(x^2-y^2)}{(x^2+y^2)^2} \quad \text{— ANS}$$

Find $f_x(x,y)$ & $f_y(x,y)$

Question 31

$$f(x,y) = \sqrt{3x^5y - 7x^3y}$$

Wrt X

$$f_x(x,y) = \frac{1}{2} \cdot \frac{1}{\sqrt{3x^5y-7x^3y}} \cdot (15x^4y - 21x^2y)$$

$$= \frac{3x^2y(5x^2-7)}{2\sqrt{3x^5y-7x^3y}}$$

$$= \frac{3x^2y(5x^2-7)(3x^5y-7x^3y)^{1/2}}{2} \quad \text{— ANS}$$

WRT Y

$$f_y(x,y) = \frac{1}{2\sqrt{3x^5y-7x^3y}} \cdot (3x^5-7x^3)$$

$$= \frac{3x^5-7x^3}{2\sqrt{3x^5y-7x^3y}}$$

$$= \frac{x^3}{2}(3x^2-7)(3x^5y-7x^3y)^{-1/2} \quad \text{— ANS}$$

Question 32

$$f(x,y) = \frac{x+y}{x-y}$$

Wrt x

$$f_x(x,y) = \frac{(x-y)\,\partial/\partial x(x+y) - (x+y)\,\partial/\partial x(x-y)}{(x-y)^2}$$

$$= \frac{(x-y) - (x+y)}{(x-y)^2}$$

$$= \frac{x-y-x-y}{(x-y)^2}$$

$$= -\frac{2y}{(x-y)^2} \quad \text{— ANS}$$

Wrt y

$$f_y(x,y) = \frac{(x-y)\,\partial/\partial y(x+y) - (x+y)\,\partial/\partial y(x-y)}{(x-y)^2}$$

$$= \frac{(x-y)(1) - (x+y)(-1)}{(x-y)^2}$$

$$= \frac{x-y+x+y}{(x-y)^2}$$

$$= \frac{2x}{(x-y)^2} \quad \text{— ANS}$$

Question 33

$$f(x,y) = y^{-3/2}\tan^{-1}(x/y)$$

Wrt x

$$f_x(x,y) = \frac{y^{-3/2}}{1+\left(\frac{x}{y}\right)^2} \cdot \left(\frac{1}{y}\right)$$

$$= \frac{y^{-3/2}}{1+\frac{x^2}{y^2}} \cdot \frac{1}{y}$$

$$= \frac{y^{-3/2}}{y^2+x^2} \cdot \frac{y^2}{y}$$

$$= \frac{y^{-3/2}}{y^2+x^2} \cdot y$$

$$= \frac{y^{-1/2}}{y^2+x^2} \quad \text{— ANS}$$

Wrt Y

$$f_y(x,y) = y^{-3/2}\, d/dy \tan^{-1}(x/y) + \tan^{-1}(x/y)\, d/dy\,(y^{-3/2})$$

$$= \frac{y^{-3/2}}{1+\frac{x^2}{y^2}} \cdot \frac{-x}{y^2} + \tan^{-1}(x/y) \cdot \frac{-3}{2} y^{-5/2}$$

$$= \frac{y^{-3/2} \cdot -\cancel{y^2}\, x}{y^2+x^2\, \cancel{y^2}} + \tan^{-1}\left(\frac{x}{y}\right) \cdot \frac{-3}{2} y^{-5/2}$$

$$= \frac{xy^{-3/2}}{y^2+x^2} - \frac{3}{2} y^{-5/2} \tan^{-1}\left(\frac{x}{y}\right) \quad \text{— ANS}$$

Question 34

$$f(x,y) = x^3 e^{-y} + y^3 \sec\sqrt{x}$$

Wrt X

$$f_x(x,y) = 3x^2 e^{-y} + y^3 \sec\sqrt{x} \tan\sqrt{x} \cdot \frac{1}{2} x^{-1/2}$$

$$= 3x^2 \cancel{y} e^{-y} + \frac{1}{2} y^3 x^{-1/2} \sec\sqrt{x} \tan\sqrt{x} \quad \text{— ANS}$$

Wrt Y

$$f_y(x,y) = x^3 e^{-y}(-\cancel{y}1) + 3y^2 \sec\sqrt{x}$$

$$= -x^3 \cancel{y} e^{-y} + 3y^2 \sec\sqrt{x}$$

$$= -x^3 e^{-y} + 3y^2 \sec\sqrt{x} \quad \text{— ANS}$$

Find $f(x), f(y), f(z)$

Question 43

$$f(x,y,z) = z \ln(x^2 y \cos z)$$

Wrt x

$$f_x = \frac{z}{(x^2 y \cos z)} \cdot 2xy\cos z$$

$$= \frac{2z}{x} \text{ — ANS}$$

Wrt y

$$f_y = \frac{z}{(x^2 y \cos z)} \cdot x^2 \cos z$$

$$= \frac{z}{y} \text{ — Ans}$$

Wrt z

$$f_z = z \, \partial/\partial z \ln(x^2 y \cos z) + \ln(x^2 y \cos z) \, \partial/\partial z \, z$$

$$= \frac{z}{x^2 y \cos z} \cdot (-x^2 y \sin z) + \ln(x^2 y \cos z)$$

$$= -z \cdot \frac{\sin z}{\cos z} + \ln(x^2 y \cos z)$$

$$= -z\tan z + \ln(x^2 y \cos z)$$

$$= \ln(x^2 y \cos z) - z\tan z \text{ — Ans}$$

Question 44

$$f(x,y,z) = y^{-3/2} \sec\left(\frac{xz}{y}\right)$$

Wrt X

$$f_x = y^{-3/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right)\left(\frac{z}{y}\right)$$

$$= zy^{-5/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right) \text{ — ANS}$$

WRT Y

$$f_y = y^{-3/2}\, \partial/\partial y \sec\left(\frac{xz}{y}\right) + \sec\left(\frac{xz}{y}\right) \partial/\partial y\; y^{-3/2}$$

$$= y^{-3/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right)\left(\frac{-xz}{y^2}\right) + \sec\left(\frac{xz}{y}\right)\left(-\frac{3}{2}y^{-5/2}\right)$$

$$= -xzy^{-7/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right) - \frac{3}{2}y^{-5/2} \sec\left(\frac{xz}{y}\right)$$

— ANS

WRT Z

$$f_z = y^{-3/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right)\left(\frac{x}{y}\right)$$

$$= xy^{-5/2} \sec\left(\frac{xz}{y}\right) \tan\left(\frac{xz}{y}\right) \text{ — ANS}$$

Question 45

$$f(x,y,z) = \tan^{-1}\left(\frac{1}{xy^2z^3}\right)$$

WRT X

$$\cancel{f_x = 1+(xy^2z^3)^{2}\; \partial/\partial x\, (xy^2z^3)^{-1}}$$

$$f_x = \frac{1}{1+\frac{1}{(xy^2z^3)^2}}\; \partial/\partial x\; \frac{x^{-1}}{y^2z^3}$$

$$= \frac{1}{\frac{x^2y^4z^6+1}{x^2y^4z^6}} \cdot -\frac{1}{x^2y^2z^3}$$

$$= \frac{x^2y^4z^6}{x^2y^4z^6+1} \cdot \frac{1}{x^2y^2z^3}$$

$$= -\frac{y^2z^3}{x^2y^4z^6+1} \text{ — ANS}$$

WRT Y

$$f_y = \frac{1}{1+\frac{1}{(x^2y^2z^3)^2}} \cdot \partial/\partial y \; \frac{y^{-2}}{xz^3}$$

$$= \frac{x^2y^4z^6}{1+(x^2y^4z^6)} \cdot -\frac{2}{xy^3z^3}$$

$$= -\frac{2xyz^3}{1+(x^2y^4z^6)} \text{ — ANS}$$

WRT Z

$$f_z = \frac{1}{1+\frac{1}{(xy^2z^3)^2}} \; \partial/\partial z \; \frac{z^{-3}}{xy^2}$$

$$= \frac{x^2y^4z^6}{1+x^2y^4z^6} \cdot -\frac{3}{xy^2z^4}$$

$$= -\frac{3xy^2z^2}{1+x^2y^4z^6} \text{ — ANS.}$$

Find $\frac{\partial w}{\partial x}$, $\frac{\partial w}{\partial y}$, $\frac{\partial w}{\partial z}$

Question 47

$$w = ye^z \sin xz$$

WRT X

$$\frac{\partial w}{\partial x} = ye^z \cos xz \cdot z$$

$$= z ye^z \cos xz \quad \text{— ANS}$$

WRT Y

$$\frac{\partial w}{\partial y} = e^z \sin xz \quad \text{— ANS}$$

WRT Z

$$\frac{\partial w}{\partial z} = ye^z \, \partial/\partial z \sin xz + \sin xz \, \partial/\partial z \, ye^z$$

$$= ye^z \cos xz \cdot x + \sin xz \, ye^z$$

$$= xye^z \cos xz + \sin xz \, ye^z$$

$$= ye^z (x \cos xz + \sin xz) \quad \text{— ANS}$$

Question 48

$$w = \frac{x^2 - y^2}{y^2 + z^2}$$

WRT X

$$\frac{\partial w}{\partial x} = \frac{2x}{y^2 + z^2} \quad \text{— ANS}$$

WRT Y

$$\frac{\partial w}{\partial y} = \frac{(y^2+z^2)\,\partial/\partial y\,(x^2-y^2) - (x^2-y^2)\,\partial/\partial y\,(y^2+z^2)}{(y^2+z^2)^2}$$

$$= \frac{(y^2+z^2)(-2y) - (x^2-y^2)(2y)}{(y^2+z^2)^2}$$

$$= \frac{\cancel{-2y^3} - 2yz^2 - 2yx^2 \cancel{+2y^3}}{(y^2+z^2)^2}$$

$$= \frac{-2y\,(z^2+x^2)}{(y^2+z^2)} \text{ — ANS}$$

WRT Z

$$\frac{\partial w}{\partial z} = (x^2-y^2)\,\partial/\partial z\,(y^2+z^2)^{-1}$$

$$= (x^2-y^2) - (y^2+z^2)^{-2} \cdot (2z)$$

$$= \frac{-2z\,(x^2-y^2)}{(y^2+z^2)^2} \text{ — ANS}$$

Question 49

$$w = \sqrt{x^2+y^2+z^2}$$

WRT X

$$\frac{\partial w}{\partial x} = \frac{\cancel{2}x}{\cancel{2}\sqrt{x^2+y^2+z^2}}$$

$$= \frac{x}{\sqrt{x^2+y^2+z^2}} \text{ — ANS}$$

WRT Y

$$\frac{\partial w}{\partial y} = \frac{\cancel{2}y}{\cancel{2}\sqrt{x^2+y^2+z^2}}$$

$$= \frac{y}{\sqrt{x^2+y^2+z^2}} \text{ — ANS}$$

WRT Z

$$= \frac{z}{\sqrt{x^2+y^2+z^2}} \text{ — ANS}$$

Confirm that mixed Second order partial derivatives of f are Same

Question 85

$$f(x,y) = 4x^2 - 8xy^4 + 7y^5 - 3$$

WRT X

$$f(x) = 8x - \cancel{32xy}\ 8y^4$$
$$f_{xy} = -32y^3$$

WRT Y

$$f_y = -32xy^3 + 35y^4$$
$$f_{yx} = -32y^3$$

$$f_{xy} = f_{yx} = -32y^3$$ ANS (Proved)

Question 86

$$f(x,y) = \sqrt{x^2+y^2}$$

WRT X

$$f_x = \frac{\cancel{2}x}{\cancel{2}\sqrt{x^2+y^2}}$$
$$= \frac{x}{\sqrt{x^2+y^2}}$$

$$f_{xy} = x\ \partial/\partial y\ (x^2+y^2)^{-1/2}$$
$$= x\left(-\frac{1}{\cancel{2}}(x^2+y^2)^{-3/2}\right)(\cancel{2}y)$$
$$= -xy\ (x^2+y^2)^{-3/2}$$

WRT Y

$$f_y = \frac{y}{\sqrt{x^2+y^2}}$$

$f_{yx} = y \cdot \partial/\partial x \, (x^2+y^2)^{-1/2}$

$= y\left(-\frac{1}{2}(x^2+y^2)^{-3/2}\right)(2x)$

$= -xy\,(x^2+y^2)^{-3/2}$

$f_{xy} = f_{yx} = -xy(x^2+y^2)^{-3/2}$ ANS (Proved)

Question 87

$f(x,y) = e^x \cos y$

WRT X

$f_x = e^x \cos y$

$f_{xy} = -e^x \sin y$

WRT Y

$f_y = -e^x \sin y$

$f_{yx} = -e^x \sin y$

$f_{xy} = f_{yx} = -e^x \sin y$ ANS (Proved)

Question 89

$f(x,y) = \ln(4x-5y)$

WRT X

$f(x) = \frac{1}{(4x-5y)} \cdot 4$

$f_{xy} = 4\,\partial/\partial y\,(4x-5y)^{-1}$

$= 4(4x-5y)^{-2} \cdot 5$

$= -20(4x-5y)^{-2}$

$= \frac{20}{(4x-5y)^{2}}$

WRT Y

$$f_y = \frac{1}{(4x-5y)} \cdot 5$$

$$f_{yx} = 5\,\partial/\partial x\,(4x-5y)^{-1}$$

$$= 5\left(-(4x-5y)^{-2}\right) \cdot 4$$

$$= \frac{20}{(4x-5y)^2}$$

$$f_{xy} = f_{yx} = \frac{20}{(4x-5y)^2}$$ — ANS (Proved)

SHow that function satifies laplace's equation

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$

Question 101

a) $z = x^2 - y^2 + 2xy$

WRT X

$$\frac{\partial z}{\partial x} = 2x + 2y$$

$$\frac{\partial^2 z}{\partial x^2} = 2$$

WRT Y

$$\frac{\partial z}{\partial y} = -2y + 2x$$

$$\frac{\partial^2 z}{\partial y} = -2$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 2 + (-2) = 0$$

— True

b) $z = e^x \sin y + e^y \cos x$

WRT x

$$\frac{\partial z}{\partial x} = e^x \sin y \cdot - e^y \sin x$$

$$\frac{\partial^2 z}{\partial y^2} = e^x \sin y - e^y \cos x$$

WRT y

$$\frac{\partial z}{\partial y} = e^x \cos y + e^y \cos x$$

$$\frac{\partial^2 z}{\partial y^2} = -e^x \sin y + e^y \cos x$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \cancel{e^x \sin y} - \cancel{e^y \cos x} + \cancel{e^y \cos x} - \cancel{e^x \sin y} = 0$$

— True

c) $z = \ln(x^2+y^2) + 2\tan^{-1}(y/x)$

WRT x

$$\frac{\partial z}{\partial x} = \frac{2x}{x^2+y^2} + \frac{2}{1+\frac{y^2}{x^2}}(-y/x^2)$$

$$= \frac{2x}{x^2+y^2} - \frac{2\cancel{x^2}\,\frac{x^2}{\cancel{x^2}}\,y}{x^2+y^2}$$

$$= \frac{2x - 2y}{x^2+y^2}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{(x^2+y^2)\,\partial/\partial x\,(2x-2y) - (2x-2y)\,\partial/\partial x\,(x^2+y^2)}{(x^2+y^2)}$$

$$= \frac{2(x^2+y^2) - 2(x-y)(2x)}{(x^2+y^2)^2}$$

$$= \frac{2\left((x^2+y^2) - 2x^2 + 2xy\right)}{(x^2+y^2)}$$

$$= \frac{2(y^2 - x^2 + 2xy)}{(x^2+y^2)}$$

$$= \frac{-2(x^2 - y^2 - 2xy)}{(x^2+y^2)}$$

WRT Y

$$\frac{\partial z}{\partial y} = \frac{2y}{(x^2+y^2)} + \frac{2y^{2}}{x^2+y^2} \cdot \left(\frac{1}{x}\right)$$

$$= \cancel{\frac{2y + 2y^2 x^{-1}}{(x^2+y^2)}} \; \frac{1}{y}$$

$$= \frac{2y + 2x}{(x^2+y^2)}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{(x^2+y^2)\,\partial/\partial y\,(2y+2x) - (2y+2x)\,\partial/\partial y\,(x^2+y^2)}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2)(2) - (2)(y+x)(2y)}{(x^2+y^2)^2}$$

$$= \frac{-2(2y^2 + \cancel{2}x^2 - 2x^2 - 2xy)}{(x^2+y^2)^2}$$

$$= \frac{-2(\cancel{2}y^2 - 2xy - x^2)}{(x^2+y^2)^2}$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = -2\left(\frac{\cancel{x^2} + \cancel{y^2} - \cancel{2xy} + \cancel{2xy} - \cancel{y^2} + \cancel{x^2}}{(x^2+y^2)^2}\right)$$

$$= 0$$

TRUE

Exercise 13.5

Chain rule

If y is differentiable function of x and x is differentiable function of t, then the chain rule for function of one variable states that, under composition y becomes a differentiable function of t with

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$$

Example 1

$$z = x^2 y$$
$$x = t^2$$
$$y = t^3$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$\frac{dz}{dt} = (2xy)(2t) + (x^2)(3t^2)$$

$$= (2t^5)(2t) + (t^4)(3t^2)$$

$$= 4t^6 + 3t^6$$

$$= 7t^6 \text{ — ANS}$$

Example 2

$$w = \sqrt{x^2+y^2+z^2}$$
$$x = \cos\theta$$
$$y = \sin\theta$$
$$z = \tan\theta$$

$$\frac{dw}{d\theta} = \frac{\partial w}{\partial x}\frac{dx}{d\theta} + \frac{\partial w}{\partial y}\frac{dy}{d\theta} + \frac{\partial w}{\partial z}\frac{dz}{d\theta}$$

$$= \frac{1}{2}(x^2+y^2+z^2)(2x)(-\sin\theta) + \frac{1}{2}(x^2+y^2+z^2)^{1/2}(2y)(\cos\theta)$$

$$+ \frac{1}{2}(x^2+y^2+z^2)^{-1/2}(2z)(\sec^2\theta)$$

$$= \frac{1}{2}\left(\cos^2 x^2 + y^2 + z^2\right)^{-1/2}\left(-2x\sin\theta + 2y\cos\theta + 2z(\sec^2\theta)\right)$$

$$= \frac{1}{2}\left(\cos^2\theta + \sin^2\theta + \tan^2\theta\right)^{-1/2}\left(-2\cancel{\cos\theta\sin\theta} + \cancel{2\sin\theta\cos\theta} + 2(\tan\theta)(\sec^2\theta)\right)$$

$$= \frac{1}{2}\left(1 + \tan^2\theta\right)^{-1/2}\left(2\tan\theta\sec^2\theta\right)$$

$$= \frac{1}{\cancel{2}}\left(\sec^2\theta\right)^{-1/2}\left(\cancel{2}\tan\theta\sec^2\theta\right)$$

$$= \tan\theta\sqrt{\sec^2\theta} \quad \text{— ANS}$$

Example 3

$$z = e^{xy} \qquad \text{find } \frac{\partial z}{\partial u} \text{ \& } \frac{\partial z}{\partial v}$$

$$x = 2u + v$$

$$y = u/v$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$$

$$= (e^{xy}.y)(2) + (e^{xy}.x)(1/v)$$

$$= 2ye^{xy} + \frac{x}{v}e^{xy}$$

$$= e^{xy}\left(2y + \frac{x}{v}\right)$$

$$= e^{(2u+v)(u/v)}\left(2\frac{u}{v} + \frac{2u+v}{v}\right)$$

$$= \left(\frac{2u + 2u + v}{v}\right)e^{(2u+v)(u/v)}$$

$$= \left(\frac{4u}{v} + 1\right)e^{(2u+v)(u/v)} \quad \text{— ANS}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

$$= (ye^{xy})(1) + (xe^{xy})\left(-\frac{u}{v^2}\right)$$

$$= \frac{u}{v} e^{(2u+v)(u/v)} - (2u+v)\left(e^{(2u+v)(u/v)}\right)\left(\frac{u}{v^2}\right)$$

$$= e^{(2u+v)(u/v)}\left(\frac{u}{v} - \frac{2u^2+vu}{v^2}\right)$$

$$= e^{(2u+v)(u/v)}\left(\frac{uv - 2u^2 - \cancel{uv}}{v^2}\right)$$

$$= e^{(2u+v)(u/v)}\left(-\frac{2u^2}{v^2}\right) \quad \text{— ANS}$$

Example 4

$$w = e^{xyz}$$
$$x = 3u+v$$
$$y = 3u-v$$
$$z = u^2v$$

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial u}$$

$$= (yze^{xyz})(3) + (xze^{xyz})(3) + (xye^{xyz})(2uv)$$

$$= 3yze^{xyz} + 3xze^{xyz} + 2uvxye^{xyz}$$

$$= e^{xyz}(3yz + 3xz + 2uvxy)$$

$$= e^{(3u+v)(3u-v)(u^2v)}\left(3(3u-v)(u^2v) + 3(3u+v)(u^2v) + 2uv(3u+v)(3u-v)\right)$$

— ANS

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial v}$$

$$= (yze^{xyz})(1) + (xze^{xyz})(-1) + (xye^{xyz})(u^2)$$

$$= e^{xyz}(yz - xz + xyu^2)$$

$$= e^{(3u+v)(3u-v)(u^2v)}\left((3u-v)(u^2v) - (3u+v)(u^2v) + (3u+v)(3u-v)(u^2\right.$$

Example 5

$$w = x^2 + y^2 - z^2$$
$$x = \rho \sin\phi \cos\theta$$
$$y = \rho \sin\phi \sin\theta$$
$$z = \rho \cos\phi$$

$$\frac{\partial w}{\partial \rho} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial \rho} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial \rho} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial \rho}$$
$$= (2x)(\sin\phi\cos\theta) + (2y)(\sin\phi\sin\theta) + (-2z)(\rho\cos\phi)$$
$$= 2(\rho\sin\phi\cos\theta)(\sin\phi\cos\theta) + 2(\rho\sin\phi\sin\theta)(\sin\phi\sin\theta)$$
$$- 2(\rho\cos\phi)(\cos\phi)$$
$$= 2(\rho\sin^2\phi\cos^2\theta) + 2(\rho\sin^2\phi\sin^2\theta) - 2(\rho\cos^2\phi)$$
$$= 2\rho\sin^2\phi(\cos^2\theta + \sin^2\theta) - 2(\rho\cos^2\phi)$$
$$= 2\rho\sin^2\phi - 2\rho\cos^2\phi$$
$$= 2\rho(\sin^2\phi - \cos^2\phi)$$
$$= -2\rho\cos 2\phi \text{ — ANS}$$

$$\frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial \theta} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial \theta}$$
$$= (2x)(-\rho\sin\phi\sin\theta) + (2y)(\rho\sin\phi\sin\theta) - (2z)($$
$$= -2(\rho\sin\phi\cos\theta)(\rho\sin\phi\sin\theta) + 2y(\rho\sin\phi\sin\theta)(\rho\sin\phi\cos\theta)$$

$$= 0 \text{ — ANS}$$

Example 6

$$w = xy + yz$$
$$y = \sin x$$
$$z = e^x$$

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial x} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial x} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial x}$$
$$= y + (x+z)(\cos x) + (y)(e^x)$$
$$= \sin x + (x + e^x)(\cos x) + e^x \sin x \text{ — ANS}$$

Example 7

$$x^3 + y^3 x - 6 = 0$$

$$\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y}$$

$$= -\frac{3x^2 + y^3}{3y^2 x} \text{ — ANS}$$

Example 8

$x^2 + y^2 + z^2 = 1$ find $\partial z/\partial x$ & $\partial z/\partial y$ at point $\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$

$$\frac{\partial z}{\partial x} = -\frac{\partial f/\partial x}{\partial f/\partial z}$$

$$= -\frac{2x}{2z}$$

$$= -\frac{x}{z}$$

$$\frac{\partial z}{\partial y} = -\frac{\partial f/\partial y}{\partial f/\partial z}$$

$$= -\frac{y}{z}$$

$$\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$$

$$\frac{\partial z}{\partial x} = -\frac{(2/3)}{(2/3)} = -1 \text{ — ANS}$$

$$\frac{\partial z}{\partial y} = -\frac{(1/3)}{(2/3)} = -\frac{1}{2} \text{ — ANS}$$

Use appropiate form of chain rule to find $\partial z/\partial t$

Question 1

$$z = 3x^2y^3$$
$$x = t^4$$
$$y = t^2$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$
$$= (6xy^3)(4t^3) + (9x^2y^2)(2t)$$
$$= (6t^{10})(4t^3) + (9t^8t^4)(2t)$$
$$= 24t^{13} + 18t^{13}$$
$$= 42t^{13} \quad \text{— ANS}$$

Question 2

$$z = \ln(2x^2+y)$$
$$x = \sqrt{t}$$
$$y = t^{2/3}$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$
$$= \left(\frac{4x}{2x^2+y}\right)\left(\frac{1}{2}t^{-1/2}\right) + \left(\frac{1}{2x^2+y}\right)\left(\frac{2}{3}t^{-1/3}\right)$$
$$= \frac{2t^{-1/2}\sqrt{t}}{2t+t^{2/3}} + \frac{2}{3}\left(\frac{t^{-1/3}}{2t+t^{2/3}}\right)$$
$$= \frac{2}{2t+t^{2/3}} + \frac{4t^{-1/3}}{3}\left(\frac{2}{2t+t^{2/3}}\right)$$
$$= \frac{2}{2t+t^{2/3}}\left(1 + \frac{t^{-1/3}}{3}\right)$$
$$= \frac{2}{2t+t^{2/3}}\left(\frac{3+t^{-1/3}}{3}\right)$$
$$= \frac{2(3+t^{-1/3})}{3(2t+t^{2/3})} \quad \text{— ANS}$$

Question 3

$z = 3\cos x - \sin xy$

$x = 1/t$

$y = 3t$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$\cancel{3(-\sin x)(-t^{-2}) + (-\cos xy)(3t)}$$

$$= \Big((-3\sin x) - (\cos xy)(y)\Big)(-t^{-2}) + (-\cos x^2 y)(3)$$

$$= \left(3\sin x + \cos xy^2\right)(t^{-2}) - 3\cos x^2 y$$

$$= 3t^{-2}\sin x + t^{-2}\cos xy^2 - 3\cos x^2 y$$

$$= 3t^{-2}\sin 1/t + t^{-2}\cos(1/t)(9t^2) - 3\cos(1/t^2)(3t)$$

$$= 3t^{-2}\sin 1/t + 9\cos 1/t - 9\cos 1/t$$

$$= 3t^{-2}\sin 1/t$$

Question 4

$\cancel{z = 3\cos x - \sin xy}$

$\cancel{x = 1/t}$

$\cancel{y = 3t}$

$$\cancel{\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}}$$

$z = \sqrt{1+x-2xy^4}$

$x = \ln t$

$y = t$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$= \frac{1-2y^4}{\sqrt{1+x-2xy^4}}\left(\frac{1}{2t}\right) - \frac{8xy^3}{\sqrt{1+x-2xy^4}}$$

$$= \frac{1-2y^4}{2t\sqrt{1+x-2xy^4}} - \frac{8xy^3}{2\sqrt{1+x-2xy^4}}$$

$$= \frac{1}{\sqrt{1+x-2xy^4}}\left(\frac{1-2y^4}{2t} - 8xy^3\right)$$

$$= \frac{1}{2\sqrt{1+x-2xy^4}}\left(\frac{1-2y^4-8xy^3t}{t}\right)$$

$$= \frac{1-2t^4-8t^4\ln t}{2\sqrt{1+\ln t-2t^4\ln t}} \quad \text{ANS.}$$

Use appropriate form of chain rule to find $\partial w/\partial t$

Question 7

$w = 5x^2y^3z^4$

$x = t^2$

$y = t^3$

$z = t^5$

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}$$

$$= (10xy^3z^4)(2t) + (15x^2y^2z^4)(3t^2) + (20x^2y^3z^3)(5t^4)$$

$$= (10t^2t^9t^{20})(2t) + (15t^4t^6t^{20})(3t^2) + (20t^4t^9t^{15})(5t^4)$$

$$= (20t^{32}) + 45t^{32} + 100t^{32}$$

$$= 165t^{32}$$ — ANS

Question 8

$w = \ln(3x^2 - 2y + 4z^3)$

$x = t^{1/2}$

$y = t^{2/3}$

$z = t^{-2}$

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}$$

$$= \frac{6x}{(3x^2-2y+4z^3)}\left(\frac{1}{2\sqrt{t}}\right) - \frac{2}{(3x^2-2y+4z^3)}\left(\frac{2}{3}t^{-1/3}\right) + \frac{12z^2}{3x^2-2y+4z^3}\,2t^{-3}$$

$$= \frac{6\cancel{\sqrt{t}}}{3t-2t^{2/3}+4t^{-6}}\left(\frac{1}{2\cancel{\sqrt{t}}}\right) - \frac{4t^{-1/3}}{3(3t-2t^{2/3}+4t^{-6})} - \frac{12t^{-4}}{3t-2t^{2/3}+4t^{-6}}(2t^{-3})$$

$$\frac{1}{3t-2t^{2/3}+4t^{-6}}\left[3 - \frac{4t^{-1/3}}{3} - 24t^{-7}\right] \text{ — ANS}$$

Question 9

$$w = 5\cos xy - \sin xz$$
$$x = 1/t$$
$$y = t$$
$$z = t^3$$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}$$

$$= (-5y\sin xy - z(\cos xz))(-t^{-2}) + (-5x\sin xy)(1) + (-x\cos xz)(3t^2)$$

$$= (5y\sin xy + z\cos xz)(t^{-2}) - 5x\sin xy - x\cos xz\,(3t^2)$$

$$= (5t\sin + t^3\cos t^2)(t^{-2}) - 5/t\,\sin - (1/t\cos t^2)(3t^2)$$

$$= \cancel{5t^{-1}\sin} + t\cos t^2 - \cancel{5(1/t)\sin} - 3t\cos t^2$$

$$= \cancel{\sin} - 2t\cos t^2 \text{ — ANS}$$

Question 10

$$w = \sqrt{1+x-2yz^4x}$$
$$x = \ln t$$
$$y = t$$
$$z = 4t$$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}$$

$$= \left(\frac{1-2yz^4}{2\sqrt{1+x-2yz^4x}}\right)\left(\frac{1}{t}\right) + \left(\frac{-2z^4x}{2\sqrt{1+x-2yz^4x}}\right) + \left(\frac{-8yz^3x}{2\sqrt{1+x-2yz^4x}}\right)(4)$$

$$= \frac{1}{2\sqrt{1+x-2yz^4x}}\left(\frac{1-2yz^4}{t} - 2xz^4 - 32yz^3x\right)$$

$$= \frac{1}{2(1+\ln t-2(t)(4t)^4\ln t)^{1/2}}\left(\frac{1-2t(4t)^4}{t} - 2\ln t(4t)^4 - 32(t)(4t)^3(\ln t)\right)$$

$$= \frac{1}{2\sqrt{1+\ln t-512t^5\ln t}}\left(\frac{1-512t^5-512t^5\ln t-2048t^5\ln t}{t}\right)$$

$$= \frac{1-512t^5-2560t^5\ln t}{2t\sqrt{1+\ln t-512t^5\ln t}} \quad \text{— ANS}$$

Use appropriate forms of chain rule to find derivative of.

Question 27

$$z = \ln(x^2+1) \quad \text{find} = \frac{\partial z}{\partial r} \ \&\ \frac{\partial z}{\partial \theta}$$

$$x = r\cos\theta$$

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r}$$

$$= \left(\frac{1}{x^2+1}\right)(2x)(\cos\theta)$$

$$= \frac{2x\cos\theta}{x^2+1}$$

$$= \frac{2r\cos^2\theta}{r^2\cos^2\theta+1} \quad \text{— ANS}$$

$$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial \theta}$$

$$= \left(\frac{2x}{x^2+1}\right)(-r\sin\theta)$$

$$= -\frac{2xr\sin\theta}{x^2+1}$$

$$= -\frac{2r^2\cos\theta\sin\theta}{r^2\cos^2\theta+1} \quad \text{— ANS}$$

Question 28

$u = rs^2 \ln t$ Find $\frac{\partial u}{\partial x}$ & $\frac{\partial u}{\partial y}$

$r = x^2$

$s = 4y+1$

$t = xy^3$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial x}$$

$$= (s^2 \ln t)(2x) + \left(\frac{rs^2}{t}\right)(y^3)$$

$$= 2xs^2 \ln t + \frac{rs^2y^3}{t}$$

$$= 2x(4y+1)^2 \ln(xy^3) + \frac{x^2(4y+1)^2 y^3}{xy^3}$$

$$= \frac{2x^2y^3(4y+1)^2 \ln(xy^3) + x^2y^3(4y+1)^2}{xy^3}$$

$$= \cancel{x}\cancel{y^3}(4y+1)^2\left(\frac{2\ln(xy^3) + 1}{\cancel{xy^3}}\right)$$

$$= x(4y+1)^2(1 + 2\ln(xy^3)) \text{ ANS}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial s}\frac{\partial s}{\partial y} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial y}$$

$$= (2rs\ln t)(4) + \left(\frac{rs^2}{t}\right)(3xy^2)$$

$$= 8rs\ln t + \frac{3xy^2rs^2}{t}$$

$$= 8x^2(4y+1)(\ln t) + 3xy^2x^2(4y+1)^2$$

$$= 8x^2 \ln \underset{xy^3}{\cancel{t}}(4y+1) + \frac{3x^3y^2(4y+1)^2}{xy^3}$$

$$= 8x^2(4y+1)\ln xy^3 + \frac{3x^2(4y+1)^2}{y} \text{ ANS}$$

Question 27

$w = 4x^2 + 4y^2 + z^2$ find $\partial w/\partial \rho$

$x = \rho \sin\phi \cos\theta$ $\partial w/\partial \phi$

$y = \rho \sin\phi \sin\theta$ $\partial w/\partial \theta$

$z = \rho \cos\phi$

$$\frac{\partial w}{\partial \rho} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial \rho} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial \rho} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial \rho}$$

$= (8x)(\sin\phi \cos\theta) + (8y)(\sin\phi \sin\theta) + (2z)(\cos\phi)$

$= 8\rho(\sin^2\phi \cos^2\theta) + (8\rho)(\sin^2\phi \sin^2\theta) + (2\rho)(\cos^2\phi)$

$= 2\rho(4\sin^2\phi \cos^2\theta + 4\sin^2\phi \sin^2\theta + \cos^2\phi)$

$= 2\rho((4\sin^2\phi)(\cos^2\theta + \sin^2\theta) + \cos^2\phi)$

$= 2\rho(4\sin^2\phi + \cos^2\phi)$ — ANS

$$\frac{\partial w}{\partial \phi} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial \phi} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial \phi} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial \phi}$$

$= (8x)(\rho \cos\phi \cos\theta) + (8y)(\rho \cos\phi \sin\theta) + (2z)(-\rho \sin\phi)$

$= 8\rho^2(\sin\phi \cos\phi \cos^2\theta) + 8\rho^2(\sin\phi \cos\phi \sin^2\theta) - 2\rho^2(\sin\phi \cos\phi)$

$= 2\rho^2(4\sin\phi \cos\phi \cos^2\theta + 4\sin\phi \cos\phi \sin^2\theta - \sin\phi \cos\phi)$

$= 2\rho^2(4\sin\phi \cos\phi - \sin\phi \cos\phi)$

$= 2\rho^2(3\sin\phi \cos\phi)$

$= 6\rho^2 \sin\phi \cos\phi$ — ANS

$$\frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial \theta} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial \theta}$$

$= (8x)(-\rho \sin\phi \sin\theta) + (8y)(\rho \sin\phi \cos\theta) + (2z)(0)$

$= -8\rho^2(\sin^2\phi \cos\theta \sin\theta) + 8\rho^2(\sin^2\phi \cos\theta \sin\theta)$

$= 0$ — ANS

Find $\partial z/\partial x$ & $\partial z/\partial y$ by implicit diffrentiation and confirm that results obtained agree with those predicted by formulas in theorem

Theorem: $-\frac{\partial f/\partial x}{\partial f/\partial z} = \frac{\partial z}{\partial x}$

Question 46

$$\ln(1+z) + xy^2 + z = 1$$

$$\frac{\partial z}{\partial x} = -\frac{\partial f/\partial x}{\partial f/\partial z}$$

$$= -\frac{y^2 \ (0+1)}{\frac{1}{1+z} + 1}$$

$$= -\frac{y^2}{\frac{1+1+z}{1+z}}$$

$$= -\frac{y^2(1+z)}{2+z} \quad \text{— ANS}$$

$$\frac{\partial z}{\partial y} = -\frac{\partial f/\partial y}{\partial f/\partial z}$$

$$= -\frac{2xy}{\frac{1}{1+z} + 1}$$

$$= -\frac{2xy}{\frac{1+1+z}{1+z}}$$

$$= -\frac{2xy(1+z)}{2+z} \quad \text{— ANS}$$

Question 47

$$ye^{x} - 5\sin 3z = 3z$$
$$= ye^{x} - 5\sin 3z - 3z$$

$$\frac{\partial z}{\partial x} = -\frac{\partial f/\partial x}{\partial f/\partial z}$$
$$= -\frac{ye^{x}}{-5\cos 3z(3) - 3}$$
$$= \frac{ye^{x}}{15\cos 3z + 3} \text{ — ANS}$$

$$\frac{\partial z}{\partial y} = -\frac{\partial f/\partial y}{\partial f/\partial z}$$
$$= \frac{e^{x}}{15\cos 3z + 3} \text{ — ANS}$$

Question 48

$$e^{xy}\cos yz - e^{yz}\sin xz + 2$$

$$\frac{\partial z}{\partial x} = -\frac{\partial f/\partial x}{\partial f/\partial z}$$
$$= -\left(\frac{e^{xy}(y)\cos yz - e^{yz}\cos xz\,(z)}{-e^{xy}\sin yz\,(y) - \{e^{yz}\cos xz\,(x) + \sin xz\,(e^{yz})(y)}\right)$$
$$= -\frac{ye^{xy}\cos yz + ze^{yz}\cos xz}{ye^{xy}\sin yz + xe^{yz}\cos xz + ye^{yz}\sin xz} \text{ — ANS}$$

$$\frac{\partial z}{\partial y} = -\frac{\partial f/\partial y}{\partial f/\partial z}$$
$$= -\left(\frac{e^{xy}(-\sin yz)(z) + (\cos yz)(e^{xy})(x) - (e^{yz})(\sin xz)(z)}{-ye^{xy}\sin yz - xe^{yz}\cos xz - ye^{yz}\sin xz}\right)$$
$$= -\frac{ze^{xy}\sin yz - xe^{xy}\cos yz + ze^{yz}\sin xz}{ye^{xy}\sin yz + xe^{yz}\cos xz + ye^{yz}\sin xz} \text{ — ANS.}$$